

Hyeonji Julia Lee

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EDUCATION

Master of Science in Information
University of Michigan

Expected May 2028

Bachelor of Science in Computer Science
Washington State University

May 2026

RESEARCH INTERESTS

Human Information Behavior • AI in Information Literacy Education • Information Retrieval Systems • User-Centered Systems and Services

RESEARCH EXPERIENCE

Research Affiliate
Center for AI Research and Education (CAIRE), Washington State University

May 2026 – Present

- Preparing manuscripts for peer-reviewed publication.

Research Assistant
Center for AI Research and Education (CAIRE), Washington State University

Oct 2024 – May 2026

- ADAPT-AI (Founding Developer): Led a developer team to build a Microsoft AI for Good-supported multi-agent assessment platform. Directed system architecture, full-stack API development, and Azure integration to support rural K–12 science teachers.
- AI Group Coordinator: Developed a full-stack, multi-agent system supported by Microsoft AI for Good to scaffold real-time scientific argumentation and promote equitable learning in middle school classrooms.
- Automated Assessment Research: Engineered an OpenAI-based pipeline for automated concept map evaluation. Validated AI scoring reliability against human rubrics using image preprocessing and voting-based data aggregation.

CONFERENCE PAPERS & PRESENTATIONS

[3] Lee, H. J., Wang, Z., Chen, Z., Liu, H., He, P., & Li, T. (2026). *Toward Development: From Customizing Classroom Assessments with ADAPT-AI to Further Supporting 3D Learning Progression*. Paper presented at National Association of Research in Science Teaching (NARST). Seattle, WA.

[2] Jin, G., Li, T., Lee, H. J., Xue, Y., He, P., Adesope, O. O., Dydasco, C. G., Sunday, O. J., & Nishida, K. (2026). *Using large language model to analyze chemistry undergraduate students' self-constructed concept maps*. Paper and poster presented at American Educational Research Association (AERA) Annual Meeting. Los Angeles, CA.

[1] Lee, H. J., & Jung, Y. (2025). *Korean and Japanese Student Flows to the U.S. and International Research Collaboration Patterns*. Paper and oral presentation delivered at the 51st Japanese Modern Association of Korea (JMAK) International Conference. Osaka, Japan.

WORKSHOPS

[2] Li, T., Krajcik, J., Akgun, S., **Lee, H. J.**, & Wang, Z. (2025). *Designing 3D Assessments with Generative AI: A Hands-On Workshop for Elementary and Middle School Educators*. Full-day workshop delivered at National Science Teacher Association (NSTA) Professional Learning Institute (PLI). Minneapolis, MN.

[1] He, P., Li, T., **Lee, H. J.**, & Wang, Z. (2025). *Designing 3D Assessments with Generative AI: A Hands-On Workshop for Elementary and Middle School Educators*. 4-hour workshop delivered at Washington Science Teacher Association (WSTA). Tacoma, WA.

PROFESSIONAL EXPERIENCE

Archival Digitization Assistant Mar 2025 – May 2026
Manuscripts, Archives & Special Collections, Washington State University

- Digitized and cataloged historical newspaper clippings (1910s–1930s) for the Kimble Project, creating precise metadata to enhance the preservation and accessibility of regional archival records.
- Facilitated researcher inquiries at the MASC front desk, providing seamless access to special collections and coordinating with specialized librarians for advanced research support.

Backend Developer May 2022 – June 2023
PCN, Seoul, Republic of Korea

- Integrated Support Platform Development: Engineered a comprehensive industry information management system using Java and MySQL, leading the project planning and end-to-end development to facilitate post-COVID industry recovery.
- Metaverse Management System Implementation: Architected and restructured a system utilizing PostgreSQL and open-source technologies, enabling administrators to seamlessly create and manage virtual metaverse spaces for hybrid exhibitions.

UNIVERSITY SERVICE

Chair, Undergraduate Student Director, Students Book Corporation Aug 2024 – Aug 2026

- Presides over board meetings to define semester goals and cultivate strategic partnerships with campus organizations (ASWSU, GPSA) to align services with the student body's needs.
- Leads the end-to-end execution of campus-wide outreach initiatives as PR Committee Chair, while managing recruitment and providing mentorship through the Search Committee to ensure seamless leadership transitions.

Industry Mentor Coordinator, Society of Women Engineers Aug 2023 – Mar 2026

- Large-Scale Event Coordination: Led the recruitment and management of STEM and architecture student clubs for KSED, overseeing safety certifications, venue logistics, and real-time operations for pre-KSED and main event execution.
- Strategic Industry & Volunteer Leadership: Managed end-to-end professional outreach to recruit mentors and industry leaders, while spearheading registration and operational committees to ensure seamless event flow for EWI and KSED initiatives.

Mentorship Program (TMP, Voiland Peer)

Aug 2023 – May 2025

- Peer Leadership & Academic Transition: Served as both a mentee and mentor within the Voiland Peer Network, providing tailored guidance to international, adult, and transfer students to ensure a successful transition to the university environment.
- Scholarship Recognition & Professional Growth: Selected as a recipient of the Boeing-sponsored TMP Mentee Scholarship, recognized for exceptional engagement and proactive professional development within the Team Mentoring Program.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, C#, TypeScript, JavaScript, SQL, HTML/CSS

Backend: FastAPI, Django REST Framework, Spring Boot, Node.js, SQLAlchemy, Pydantic

Frontend: React, Vite, Tailwind CSS, Radix UI, Shadcn/UI, React Hook Form, Figma

AI & ML: OpenAI, Multi-agent Systems, PyTorch (CNN), NLP

Cloud & DevOps: Azure, Docker, AWS, Nginx, GitHub

Database: PostgreSQL, MySQL, MariaDB, MongoDB

CERTIFICATES

Fundamentals of Deep Learning (NVIDIA Deep Learning Institute)

Generative AI for Educators & Teachers (Vanderbilt University)

NCS (National Competency Standards) based Training Course in Big Data Development Practice using JAVA & Python (Greenart)

Microsoft Office Specialist Master Certification (MOS) 2016 (Microsoft)

REFERENCES

Tingting Li, Assistant Professor (Co-Director, CAIRE Research Lab)

College of Education, Sport, and Human Sciences

Washington State University

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Peng He, Assistant Professor (Co-Director, CAIRE Research Lab)

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Joleen Warner, Digital Projects Manager

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